Functions

Learning Objectives



After learning this chapter, the students will be able to

- Understand the Definition of Functions and uses of Functions
- Understand the Types of Functions – pre-defined and user-defined functions
- Apply mathematical functions for solving problems.
- Use String and Character functions for the manipulation of String and Character data
- Implement modular programming by creating functions
- Understand the role of arguments and compare different methods of the arguments
- Recognizes the scope of variables and functions in a program.

11.1 INTRODUCTION

A large program can be split into small sub-programs (blocks) called as functions where each sub-program can perform some specific functionality. Functions reduce the size and complexity of a program, makes it easier to understand, test, and check for errors. The functions which are available by default are known as **"Built-in"** functions and user can create their own functions known as **"User-defined"** functions.

- Built-in functions – Functions which are available in C++ language standard library.
- User-defined functions – Functions created by users.

11.2 Need for Functions

To reduce size and complexity of the program we use Functions. The programmers can make use of sub programs either writing their own functions or calling them from standard library.

1. Divide and Conquer

- Complicated programs can be divided into manageable sub programs called functions.
- A programmer can focus on developing, debugging and testing individual functions.
- Many programmers can work on different functions simultaneously.

2. Reusability

- Few lines of code may be repeatedly used in different contexts. Duplication of the same code can be eliminated by using functions which improves the maintenance and reduce program size.
- Some functions can be called multiple times with different inputs.

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